

Room 8 (P1 N=6, remaining gaps): moderator summary

2026-09-26

Two seats attacked the two most concrete remaining pieces from Room 1.

Hilbert: INCONCLUSIVE, honestly and usefully. The 5 angular loci from Room 1's ray-lemma analysis cannot literally be checked, because no N=6 analogue of `prop:qi`'s $\Gamma_{\pm}$, Π_0 contours has ever been constructed in this repository – only the $N = 4$ ($q = i$) version exists. Reading the $N = 4$ template directly (rather than guessing) revealed a real structural escape pattern that might transfer: the angular ray-lemma mechanism is only invoked on bounded windows and two fixed-height infinite legs, never on the one genuinely unbounded/tilted leg, which is instead certified via an angle-free modulus bound immune to resonance loci entirely. Whether this transfers to a hypothetical $N = 6$ construction is unknown; building that construction is substantial new work, not attempted here. Confirmed explicitly: even a fully favorable resolution of this piece would *not* close $N = 6$ by itself (the $N = 6$ -specific landscape bookkeeping is still missing), and this piece is independent of Ramanujan's separately-confirmed doomed crude-method margin (different method entirely).

Erdős: a genuine small closure, reviewed and fixed. A rigorous Cauchy-estimate derivative bound on $|B'|$ now closes the specific aliasing gap Reviewer AQ flagged for the 2 certified zeros near $\zeta = e(1/6)$ from Room 1. Reviewer AT independently reproduced the computation bit-for-bit and confirmed the Cauchy-estimate mechanism, the chord quantity, and the numerical margins are all correct, with real slack (angular sweep $\approx 8^\circ$, far below the π threshold, not a knife-edge). Two write-up-level gaps (a chord-vs-arc conflation in the MVT argument; an over-stated “validated” tail bound that was really only checked over a finite window) were found and fixed in the note text; neither affected the numerical conclusion, both are now stated precisely. **Both zeros are now fully rigorously certified,**

upgrading them from “one notch below full rigor” (Room 1) to a genuine argument-principle certificate.

Room verdict: ADVANCED. A small, genuine, now-rigorous closure (2 certified zeros of B near $e(1/6)$) plus a precise clarification of why the main gap (the ray-lemma degeneracy at $\theta^* = 0$, needing an $N = 6$ -specific contour construction that does not exist) resists direct attack. **N=6’s main obstruction is unchanged and remains OPEN.** The 2 certified zeros are a minor, existence-only result (not a density/accumulation theorem), and are left repo-only for now; they could be mentioned in paper2a as a footnote-level fact if desired, but do not materially move the paper’s stated results.